

ActInf Textbook Group ~ Cohort 5 ~ Session 15

(Chapter 6, Part 2) ~ 2/12/2024

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all right welcome back everyone it's February 12th and second discussion of chapter 6 for cohort 5 so any thoughts or feelings on where to begin with chapter like any thought or quote or [Music] question I have one yes please um I'm not so sure it's in this chapter but can you go into more detail in terms of uh the difference between the high road and the low road yes would anyone like to give a thought on high road and low road in the chapter six so really getting to the recipe and the cooking what are the high road and the low road well one thought is chapter 6 is a recipe for actually walking on the low road because this is where the generative model is constructed and specified so it kind of starts from having nothing I don't know how deep we um uh have or or need to go into the food maybe it's somewhere here but I mean what are the ingredients what's the kitchen what needs to be done in what order that's the low road of building any model there could be a chapter six for a recipe for Designing SPM analysis or designing linear regression analysis as thousands of other guide books for modeling are out there so in that sense that all modeling and uh construction like this is low road because it's actually putting uh model down that's what's similar about chapter 6 and another recipe for basian modeling and then the difference with active inference is the connection through the high road to the free energy principle so that's in so relative to if I were to to translate that um and you know and I'm the model that I'm I'm building is more of a a self-modeling not trying to build a software llm model um so in terms of yeah the low the low road is is Imagining the pathway and the high road would be taking the pathway would that be a way to contrast them you know simp IC form what would someone else say Yes um Ali and then anyone else who raises their hand okay so uh the high road uh can we described kind of topdown approach to active inference which starts by asking some uh very fundamental questions about uh about what's required for any Cod UN code thing to exist and what are its necessary properties so uh by beginning from uh a kind of as friston himself would call it a near existential nihilism uh it gradually begins to build uh some framework which can account

for any sentient behavior that agent must possess in order uh in order for it to possess its integrity and to persist through time so that's basically uh the premise of high road but uh on the other hand for low road it begins by asking it it begins by the premise of uh the existence of the mind so here uh the the existence of the agent or rather the predictive agent is presupposed and then it begins to formulate or describe uh What uh kind of mechanisms uh can realistically describe uh its uh predictive behavior and it's uh I don't know brain activity uh or rather the way that the brain act the brain constructs uh the inner model or uh in know words its generative models uh so that's basically beginning from the premise of the existence of mind and uh to describe uh what kind of um mechanisms uh that particular mind should possess in order for it to uh to do both inference and action in a changing environment but ultimately they uh they achieve similar goal but uh as uh the I mean the name of these approaches would suggest uh only the approach to achieve that same goal uh is different but then again uh they basically try to uh I mean describe the same kind of behavior so you the near existential nihilism would uh would that um would that correlate with let's just assume they have agency okay so uh all right so yes agency is a kind of I mean Central presupposition for any kind of um any kind of thing that can meaningfully uh navigate through its environment and to have any kind of goal directed Behavior so in other words agency is a kind of synonym for a gold directed Behavior it can be um obviously I mean argued whether agency really equates uh gold directed Behavior but that's basically as far as I understand uh that's the meaning of agency in active inference literature or or at least in uh the majority of active inference literature thank you um Kristoff then Andrew yeah I just want to say that for me the um kind of a enlightening perspective on the on the low road um is to Simply derive the um uh an equation that features a free energy essentially from from Bas rule so you can take the Bas rule um you know uh write it in such a way that um you substitute another distribution for the um for the posterior distribution which is not not generally known um with an approximate posterior and say like okay since I cannot know the true posterior I'm going to approximate the true posterior of an approximate posterior using some kind of variational methods and essentially using just um the prod sum probabilities and rearranging those equations a little bit uh you effectively get to the point where um you have a quantity that is mathematically isomorphic to you know a free energy in thermodynamics so that that's a kind of a very mechanical mathematical way of um teasing the the concept of free energy out of essentially out of nowhere right because you start with an equation that's that's true B base um B Rule and then you make it slight of hand where you just simply substitute you know Q for p

and say like this is a slightly different kind of thing because I'm going to have a bunch of parameters that Define what that Q thing is and those parameters I can somehow tweak to find how good my uh approximate um approximate posterior is um but because for simple um mechanical um substitutions like that you can derive effectively an equation that looks like um uh free energy that's how you just get all the way from Bas fum to to the uh to the free energy principle effectively so that's that's kind of like why it's bottom up you sort of construct it um this way amazing thank you Andrew oh on me Andrew just a cute analogy I would say that Plato would always try to start with the high road while um Aristotle would uh see what he's got available and start with the low road Road and since we're trying to implement something we got to use the tools we have accessible and so um chapter six is uh seeing what we've got there and taking the low road to uh Implement something that shows active awesome yeah great great comments here's figure one two it's a nice evocative figure also though the it's not like there's a strict sequence or like there's any it's like um some subway stations but it's not very strict and it's not like all or necessary sufficient or anything like that um but it is situating active inference generative models as on one hand implementing Bas theorem giving what Kristoff um provided this kind of unrecognized relationship between statistical principles of least action which we now call the basy mechanics and the free energy principle and how that's like an extension of other physics and information and thermal physics and yet all all of that kind of like Theory aside chapter 6 doesn't have an equation chapter 6 is like actually the sequence although they um make clear like it's more about a checklist um than a than an ordered sequence here um but this is the theory um light sequence of questions that's actually procedural like a catechism to work towards doing a generative model to working in active inference and realizing the synthetic active inference agents whether it's about oneself um that's human system of Interest or whether it's about some other system um that's chapter six it's with only looks like that one figure and not any um of these connections that have been brought up which are super great and well said about the basing mechanics so chapter six on the low road but if you um build it this way then by making an active inference model from the low road it would be uh you could run it in pmtip so it would be kind of concorded with free energy principle and we have we have many examples we can look at we can look at the tutorials of pmdp we can also talk more about the recipe of chapter 6 but that's kind of the big idea of six um anyone can raise their hand with like a or or yes I have a question yes go for it um yes I wrote it in the chat what is meant by the first step the identification of boundaries and then it says I.E Markoff blanket but I just want to say thank you so much for this very um

elucidating explanation for high load and Rollo I've read it several times and this was the best or finally it it clicked so thank you thank you so great question Alie want to go for it or how does it come into play in in the process of building a generative model yes please yeah just wrote it in the chat as well so markup blanket is a kind of a statistical boundary in the state space of the agent so in other words it can uh manifest itself as a spatial temporal boundary or not so in some cases for example in U I don't know cell boundaries or cell membranes it actually manifests uh as uh real spatial temporal boundary but it's not necessarily so uh and uh it can u i mean markup blanket can be expressed variously in different forms for different situations uh but basically it kind of provide uh a necessary statistical boundary in the state space of the agent uh in order for uh for the agent to be uh statistically separated from its uh so-called environment uh so if there wasn't any Mark of blanket delineating the this boundary then uh the behavior the internal behavior of the agent uh would be identical uh to the uh behavior of uh the environment in which it's embedded so uh yeah that's a necessary presupposition for doing any kind of active inference modeling uh around that so may I ask a follow-up question yeah sure so in the case of swarms or plasmoids where an act an individual agent is impacted by the whole group um so would the same principles apply in characterizing the agent as the group yeah that's very great question so you see Mark blanket uh can be um can be described in uh multiple scales so for instance um even for um for so let's uh take the brain as an example uh we can draw a markup blanket around single neurons or around I don't know uh particular circuits neural circuits or around the whole brain and even even further than that around the whole organism and there are some Works uh that that have been used for um they've used active inference for modeling the whole um some of the social and cultural Behavior Uh of whole uh communities or societies so in those cases markup blankets can be drawn around a whole Community or Society so basically it depends on the system of interest uh it's not something predefined thank you I'll give one other example just because this swarm modeling topic um is a great one and to give a a specific example so when we were simulating an ant colony and there was a chapter 6 but this is the result of the chapter 6 this is the specification of the enviro IR m a t maze where there could be movement and then this is the generative model that was specified like the generative model of the nestmate was the ABCDE and the the kind of um active inference Loop that it implemented at each time step so in that sense it was an agent based simulation much like net logo or using any other agent based simulation except that the the the basian graph of the structure that it was implementing was like the figure 4.3 graph so specification of the environment specification of the um generative

model including the boundary in space and time like having the ability to receive um uh in a one square perimeter so we didn't a priori set a blanket on the colony but you could imagine if you said okay well I'm going to look down from the top and enclose another blanket around the total space that the colony encloses which might be meaningful or might not be but from the modeling perspective all that was needed to specify was the nestmate but then summary statistics could be made about areas covered so there's a lot of other um analogous examples with multi-agent simulations um to a first approximation what you specify is the agents just like any other agent based model um and then um people are looking into as Ali mentioned like integrating true multiscale systems modeling like multi-agent models of neurons constituting circuits and so on Andrew and then Pros yeah just a quick note that the Markoff blanket is best thought of as a computationally uh meaningful parameter not necessarily something that you will be able to pin down in physical space so for example like if you think of your Markoff blanket it's uh associated with the sensing and action um modalities you have available um you're even though there's physical space is going through you in a certain way and you are regenerating in terms of um your particles at a certain rate um it's related to how you as a computational agent are associating with your environment and so uh you can think of that both from the intrinsic perspective I just described as well as if you are modeling something you get to pick what is the data going into and being processed by your statistical model and what is the data going out and being processed uh and transformed into physical uh phenomena got it thank you thank you pros and then Kristoff yes uh thank you uh Daniel I have a question for you like in the ant model which you showed I was imagining that if we have say multiple ant and they want to form a group by themselves uh so will you be able to create like kind of a markof model like a blanket dynamically am I correct or incorrect well few things to say on that so first the as the blankets are features of the maps not the territories like it could be possible you could say again I'm taking a top down view of the movement and I'm going to draw little enclosures and I'm going to have a thresholding parameter for when I declare like these three to be a separate cluster so so but at that point that's not really like what the nestate is doing which this whole time hasn't changed from just as specified so there's a lot of higher order clustering and and partitioning methods that aren't exactly the base level specification of the generative model which is often pretty small and and simple um and then another thing to say on it is like uh as with many agent based model like here all they could do was um move and uh put down one pheromone there wasn't like an option for them to have a second pherone or to do nursing or uh other kinds of

behavior so um there's always kind of a limited zone of of the like emergence or phenomena that are displayed from a simulation if it's really pruned down to be simple and then to make um richer like more more whole iguana simulations that's what is exciting about active inference uh one one followup question on this uh so basically you are designing it through policies which you have or it's like more higher order stuff this one the only policies they could take there's no planning was just moving in a grid like a grid world but as with any active inference generative model like chapter 6 describes all the aspects of the model are specified the input sensory any um higher order beliefs that it entertains like the the cognitive States than the affordances it has how it does policy selection whether it do planning whether it has any of these other features like attention so figure 4.3 which is basically what is here um is just the minimal kernel Motif in discrete time and then there's all of these kind of like features and this is a very simple framework with the goal as most of the published models are and then um that's why there's so many opportunities now to connect the different systems of Interest with the colonel with the motifs that are arising um okay Francis yeah thank you Francis oh sorry Kristoff and then Francis then Tanner uh I just want to add one thing to the um Mark of blanket concept um slightly different perspective that uh comes from something that um Steven walr is working on the Observer Theory which is he's kind of rediscovering very similar thing to active inference but on slightly different um um basis so he talks about what's necessary for for something to be on Observer and in some sense you need to have more computation going um going on on the inside of the Observer and you know that's that's going on on the outside what what is being observed so this Mark of blanket is the statistical definition of a mark of blanket to me is is kind of um very confusing right like there's this uh statistical what does the statistical Independence really mean well it really means that you you know the the The Observer or the agent is going to do something internally that you can't readily predict from just looking at it without um it actually responding somehow to the event so it has to you know um some information is coming in it has to process it in some way and it you know it will react react in the eventually in the environment and as long as it's doing enough computation on the inside then um the uh it will basically become statistically independent because you can no longer using simple statistics or any kind of Statistics really model uh what is it really um what is it really doing uh internally and these things kind of seem to arise naturally as uh any kind of complex systems uh interact so so these are kind of boundaries that on one hand we can we can draw them somewhat arbitrarily sometimes on the other hand they're they're kind of naturally arising in interactions of any kind of um any kind

of computational systems I hope it's this is very helpful for me but um I'm not sure if it makes any sense in this context here yeah it's awesome like you're you're right they can absolutely be proposed um out of nothing like you could just draw arbitrary base graph and just say here's the blanket but they also do arise naturally seemingly um or maps with this character describe complex systems um a simple example of which being space um or time like things kind of propagating outwards in space and things not teleporting through space like a bubble expanding outwards not like skipping a shell because it's like propagating locally in space so that kind of captures that property too as well as the I guess as wol from is looking into more of the computational parts of like measuring and and observing um Francis then Tanner and Ali hi so uh my first stab at understanding marov blankets was uh from my programming background I thought okay so this is like a bit like having a function or an object where you pass parameters in which might be the goals and then leave it to get on with this and then maybe it'll some if something goes wrong it'll send an exception up to uh so that can go up the perception the exceptions go up the perception tra uh hierarchy um and then I just been I picked up the uh references to the environment and I'm starting to think of as a mark of blanket as being something which uh because things are hidden from the environment they're not part of the environment therefore they can have a is the internal State and it can be the internal state which as an to the intent that to the extent that it's an agent the internal state that it is trying to preserve um so it's kind of what's at stake what it is um at it being uh I guess anything which has a barkoff blanket um actually what what is the answer to that question what is it that can have a markof blankets uh an active entrance okay thank you thank you Tanner okay so yeah I have a I do have a question that I want to ask but first I um I want to respond to the last participants query um I think the the issue that a lot of people have when trying to understand what marov blankets and marov boundaries are is they like make it too agential in the sense of like the words that they use just become agent identified like the way that I understand it is that marov boundaries are agents right so so we say but it's important to understand that marov blankets aren't actually extent in like material reality they're just ideas right they're like this is like this is the boundary um so you know it it you can sort of understand this if you like think about it and you're like you know I so I see a spider right and I'm like that's a spider that's that's an individual but is it like you just kind of made that definition right you you assigned the you drew the Line in the Sand right you you know circled The Spider and we're like that's that's the entity right there that's the agent with identified active awareness doing the inferential you know stuff it's kind of arbitrary so marov

blankets are like defined in this uh you know more hor horis way like they are they are emergent properties that do exist like they they do they exist in a qualitative sense um you know most of the time I'm existing as a person in a body but in a state of meditation you can sort of start to exist with a larger or just different markov blanket that includes maybe other people the room Etc so that's that's all I wanted to say about that it's not it's not one to one right there's there's there's this there's a whole Spectrum to these things um anyway my question was about spiders so you uh were you have these models um that you were presenting about ants and my curiosity is uh ants have this as I understand largely two-dimensional kind of experience where it's like left right you know they're not really moving up down it's it's very much so just like follow the fairmon trail um and you can correct me if I'm wrong because you know much much more about this than I do but my my question was like if you were trying to extend that uh this model that you've created to a spider which is something that exists in in a very three-dimensional way and even in in some sense in like a four dimensional way because the whole concept of building a web is like working with a plan that is a long-term plan to catch and exist right it builds a home for itself that's not something a drone ant is doing I don't think so just curious what you kind of think about that idea if just take like a a classical like field spider as an example um and and I would I would love to to hear your thoughts on that in comparison to your model so okay just very briefly um the Markov blanket is definitely not associated with just the agent environment interface like this node is a blanket between X and Y so the markov blanket can be identified between any nodes but one kind of interest is this agentic and and definitely it it's like very um Observer dependent like all observation is how agentic different boundaries are seen as and um to the spider I mean this is what chapter 6 is about please look through chapter 6 and mega spider model and then next week let's look at it and we'll further along and maybe we can even you know go from there but that's exactly what chapter 6 is about yeah um do you have a recommendation do you have a recommendation of a particular species I should look at or I can find one I I mean from the beginning of when you're feeling curious about the phenomena on through the active inference model you make just document it and then it'll be done okay cool thank you Daniel I appreciate it thank you uh so uh little bit I think u a more sophisticated uh version of Mark of blanket I'm not sure if it's directly uh inspired by the concept of markov blanket or not may or may not be but uh there's this uh important paper by Michael 11 one of its um I believe one of the most highly cited ones the computational boundary of a cell from 2019 uh in which he develops uh the concept of scale-free cognition and the idea is that there's

this computational boundary or as um as he describes it in this paper a kind of light cone a cognitive light cone that can Define uh the scale of the agent's cognition um I mean through various scales and various limits of the cognition so it's not restricted uh to uh even a single scale of interest uh it can be uh modeled u in a in a true hierarchical way to include uh various um scales and various limitations of cognition even if we want to kind of investigate the interaction uh between uh or among um the various scales of cognition and the other uh related um conception uh was developed by uh David kraar at all and their 2020 paper the information theory of individuality in which they again uh describe a kind of U information theoretical conception of uh what it means to be a biological individual so uh this information theory of individuality uh enables uh the identification of uh of the agents at uh any level of organization I mean from uh even molecular uh molecular level to even the cultural ones uh so uh these recent con Concepts uh can be seen as kind of update on the or we can look at them as kind of Mark of blanket 2.0 but I believe uh familiarity with these newer Concepts can uh even help us to understand uh the nature of Mark of blanket uh I mean uh more clearly uh than just looking at the Active inference literature because I agree that this concept can seem a little bit vague uh for newcomers uh or even for experts as well uh so I highly recommend those two papers as well thank you Lee yeah a few comments on that like the markof blanket concept comes from the causal basian modeling literature and then there's various extensions and kind of of analogous the the the blanket index all these other things um to kind of uh I mean bring it to a more low road building does anyone want to look at the questions or share some of their modeling experiences or we could look at the PDP tutorials because the end of chapter 6 not not like it only has to be about this chapter but the end of chapter 6 um doesn't necessarily leave one um further along or scratching their head about these uh like multiscale systems uh except through the construction and often like going through and talking through and building through a system gets really far and then even the theory questions from there are more informed so it's not like the theory has to be resolved like before one builds anything okay got it that's useful like the PDP tutorials are are we or or does anyone want to look at or what would be helpful in our last bit I I think like yeah we should if everybody agrees pmdp yeah um so like let's do it there are some uh I think a model stream 7.1 and two with Connor Hines and Daphne dcus um like but those are long videos and um also contain a lot more of like the the background on pmdp um so I just put it into the six okay so um I mean the big picture on what's going to happen here and how this is different than chapter six this is not um this notebook itself is not um a supplemental dish to chapter six however with some very

slight tweaks it could be so just to kind of set set the scene like chapter six is a sequence of questions it's a catechism basically for going from your curiosity to the constructed model so what system are we modeling are we modeling one spider three spiders three spiders in a web three spiders in two webs five spiders on N webs two trees you know just that is not a trivial question what are you modeling and so figuring this out is what they have as the first question then what's the appropriate form of the generative model one big distinction being like figure 4.3 discrete time or continuous time and all the strengths and weaknesses of these different um techniques um also pmdp is like only for discrete time models right now but there's other approaches for continuous time modeling um then what's the structure of the generative model is it an estimate that can move two two squares forward or one Square the side or this way diagonally or how many pheromones and how many uh does it have a tension and if it doesn't have those features it's not going to have them if it doesn't have three-step time Horizon planning your generative model will not have that feature like if it doesn't have planning in the code it will not do planning now that doesn't mean it won't give behavior that looks very interesting in other ways but actually that's a feature because you can say through the fully synthetic nature of the model it doesn't do planning and yet this is happening um or it doesn't have a preference for this and yet this is happening and then what they leave for last is like how to set up the generative process however um as Ali and uh asked Maxwell ramstead on in the guest stream they have kind of moved away that group and of modelers not saying this is the the only the best way they've moved away from distinguishing so much between generative model and process because in the end they're co- defined and in like complex systems it's not so simple it's like which one's the agent which one's the environment it's like well there's two agents communicating so you know why not just call it all the generative model um okay so chapter six gets us to writing the pseudo code basically that we're going to look at the python code for um what they were going for here their goal was um you know if if we were to apply the chapter 6 question is to model an agent that moves in this um 3x3 maze grid world and has a preference for a certain location and you can copy the the um code this package that they have written is called PDP um and it also is bringing in just a few math packages this box this is tutorial one this box has some kind of um just purely visualization based functions like plotting likelihood this is not like an active inference box um also as some of my friends have remarked from computer science is like there's some pros and cons to notebooks on one hand they can be be copied really easily and run in the cloud like on a Chromebook but at the same time it can like lead to a lot of like okay I thought we were going

to do active inference but then no it's like this other thing is happening first so it's not always clear what's what's happening even what because it's so linearized okay here they're just showing what this function does and plotting a categorical distribution so it's it's kind of built it's like a little step by step this is not building the agent yet then from just doing this is like a three-sided coin flip now the other kind of key piece of the Bas formula is going to come into play which is conditional probabilities and they show doing some conditional probabilities just to show in PDP like what we're doing is probabilities and conditional probabilities like the The Joint distribution of the nestmate is just the joint distribution of a b c and d that totally describes the nestmate there's no other information about the nestmate and then conditional probabilities become really important because like we're often interested in conditional distributions like the distribution of action conditioned upon dis observation other things like that okay so that that was all like kind of preliminary PDP utilities and just demonstrating the syntax kind of now they're going to go into grid World okay the grid is specified it's a 3X3 grid that's specified as this um set of ordered pairs and then it's visualized with their graphing function so that's the generative process that's basically the environment is being specified now just like for other discrete time active inference agents what there is to specify is a b c and d so just to the figure uh or chapter 7 where it go which is the next chapter which goes into more detail on the discrete time model so a b c and d a b c going into um G here's C because policy distribution is conditioned upon preference and habit so it's just kind of getting wrapped into this G visualization a b c d first they generate a a is the conditional distribution that is basically mapping between observations and hidden States so um in this world each of these locations is going to um just it's basically labeled like with its location so it has like a perfect um GPS um so the dimensionality of a is set the dimensionality of a is basically like observations mapping to Hidden States it's a matrix that describes that mapping and so they're the same in this case then it's made out with zeros just to initiate it and then it's filled on the diagonal with ones which corresponds to perfect mapping of States if the a matrix were flat there'd be no information about the world coming in ETC here it's the identity Matrix so it's just a noisess mapping that's what it looks like so here white square is one likelihood black square is zero likelihood and then it's mapping from the nine observations to the nine hidden States and it looks exactly this like if you see a three it always maps to being in state three here and then um they make a noisy a with some different noise patterns so here observation zero is ambiguous whether it maps to 01 or three so following in the grand SPM tradition even very large generative models can be specified like with these grayscale heat Maps it's actually

really cool two ambiguous locations now six is ambiguous as well okay so that's a so we have a few different matrices as with other modeling it's never just about one one perfect model that describes like the spider it's about like having multiscale portfolios of matrices and composing them and displaying like like this one has this behavior and think it's cuz this but actually here's this other model that shows that it's not so it's it's not just about making one it's about like doing a lot of models and sweeping across them now they go on to b b is the hidden State transition um each slice of B is going to describe the mapping from one world State at past time point to the next time point and then different slices of B get selected based upon the um policy selection so B is the critical variable in active inference this is where the action dependent World model is specified and like this is really the variable that explodes a lot with model complexity because the D variable is just a prior on world States so it's just a vector that kicks the chain off one time with like an initial Distribution on room temperature and a is a sensory mapping from thermometer reading to the hidden State and then C is just a preference over thermometer readings so all of the work in the model with the action dependent causality all of it is in B it's basically the only place they can go just in practice so bees created these are the affordances they're called actions in this script um there are five of them up down left right stay um a few pieces are uh just visualized and just checking through that uh they can advance the next state and then um like take the action and move from here and then move right one square to here then move down to here so that's B then C they zero it out C has the same length as observations because it's a distribution over observations and here it's given a value of one for eight and zeros for the rest and then D has a length of the number of world States and it's a distribution of our world States and so here it starts 100% of the time state zero and that's just how it is so that's that's the ABCD specification that's all of the so at every single time Point all of the all that the agent simulation is one agent or multi-agent it's just a b c and d for each agent in this most simple example then they make some um functions in first state that's like the making part and this just describes the pseudo code for going from the observations combining that with a prior and returning um Q of hidden state that we discussed Q of X earlier today they do a Time step of just the sensemaking inference including with the ambiguous priors so that's the sensemaking part then action selection this is the action part um some more functions are introduced pretty cool to see the uh entropy and the K Divergence just toine so shortly they just kind of replot things and just kind of refresh where it's all at change the preference to five add a few um print statements compare the expected free energy for moving left versus moving right classical approach you have your

affordances that are being considered here just those two just for clarity in the tutorial and then EF G is calculated for those two and then a softmax is applied such that the units of G are normalized to probability distribution and then here is the whole um grid world and then they all wrap it together and then they just output how it happens and then there's a few more details and then they add planning at the end but everything above this point is no planning and then from this point you you start to see the planning I hope that's useful um and that people make also other tutorials or maybe contribute edits to PDP because there's a lot of experience with modeling that people have that that could really help and it's it's um really fun with the theory but also it's a very specific modeling possibility and we have the um there's there's so for for those with computer experience many of these will be accessible um for people with less coding experience like it will be really helpful to figure out and like learn what approaches um are exciting for you because this is the low route a low route it's a simple model but it is kind of it helps calibrate and Benchmark like if this is um a big pill then you know simulating 500 social agents with memory is probably a really challenging project for you if this is very simple and understandable then you can probably start to Benchmark what kinds of models you'll be able to understand or or at least um iterate on starting tomorrow okay any questions in the last five minutes or thoughts on like this modeling topic as that's kind of six yes prau and anyone else uh yes I was actually I think uh first thing I what I'll have to do is I'll have to just uh do this tutorial once like it's very nice you have given a brief summary of it so it'll be very helpful if you do it today or tomorrow um and I was I was thinking like if the if the Matrix is what you mentioned over here if that is like we have various concepts like various class data structures we know very well so similarly if you know this matrices quite well then I think things will just flow that's what I understand out of it right thank you yes totally and also um the the model presented here is a pretty um common PDP that's what gives the continuity with the computer science and the base graph and everything like that that it is just a PDP basically and and yet it's um about that Motif and then about understanding it in the single agent case and then just kind of building from there so like a lot of the the work and the challenge in application look forward to everyone also sharing their experiences on this is like even building one agent and then there's like the just like Create the environment with n but also people are making amazing toolkits to help facilitate multi agent simulations so just like everyone keep you know learning and exploring and the tools really are getting better fast Susan um so I'm um my head's around how to integrate this with the Constructor law and I'm wondering where um degrees of freedom show up in this

particular chapter you pointing to yes in the narrow not necessarily in the software but I in the real world oh I mean the chapter describes several of the mega degrees total degrees of freedom like okay what is the system we're modeling and who are we doing it for and why I mean these are the ultimate degrees of modular Freedom h and then these distributions have their own statistical degrees of freedom okay so it really is the construct of freedom is is kind of underlying it's kind of you know under the soil you might say of the uh of the blanket so the perception of someone's Freedom may actually correlate with where they see their boundaries where they see their Market boundaries I almost want to say say that um if I were to transfer translate the the market the markof blankets you know to a you know a okay metaphor it's not perfect but an okay metaphor it would almost be the protective boundaries that we um that we perceive in the world which has a lot to do with you know our perception of Freedom would that be however you feel all right yeah that trying to stay with the rigor um say I don't want to use something that doesn't it's completely off like I love that you added it I don't know if they talk about Freedom at all in the book so in in the letter of the book no but I really like what you added so okay thank you thank you everyone peace see you next week also feel free to do the tutorials add questions add notes and like your experiences if you were familiar with the code or not and all of this could be really fun so see you next time bye thank you everyone thank you that was great awesome bye